

all continuous,

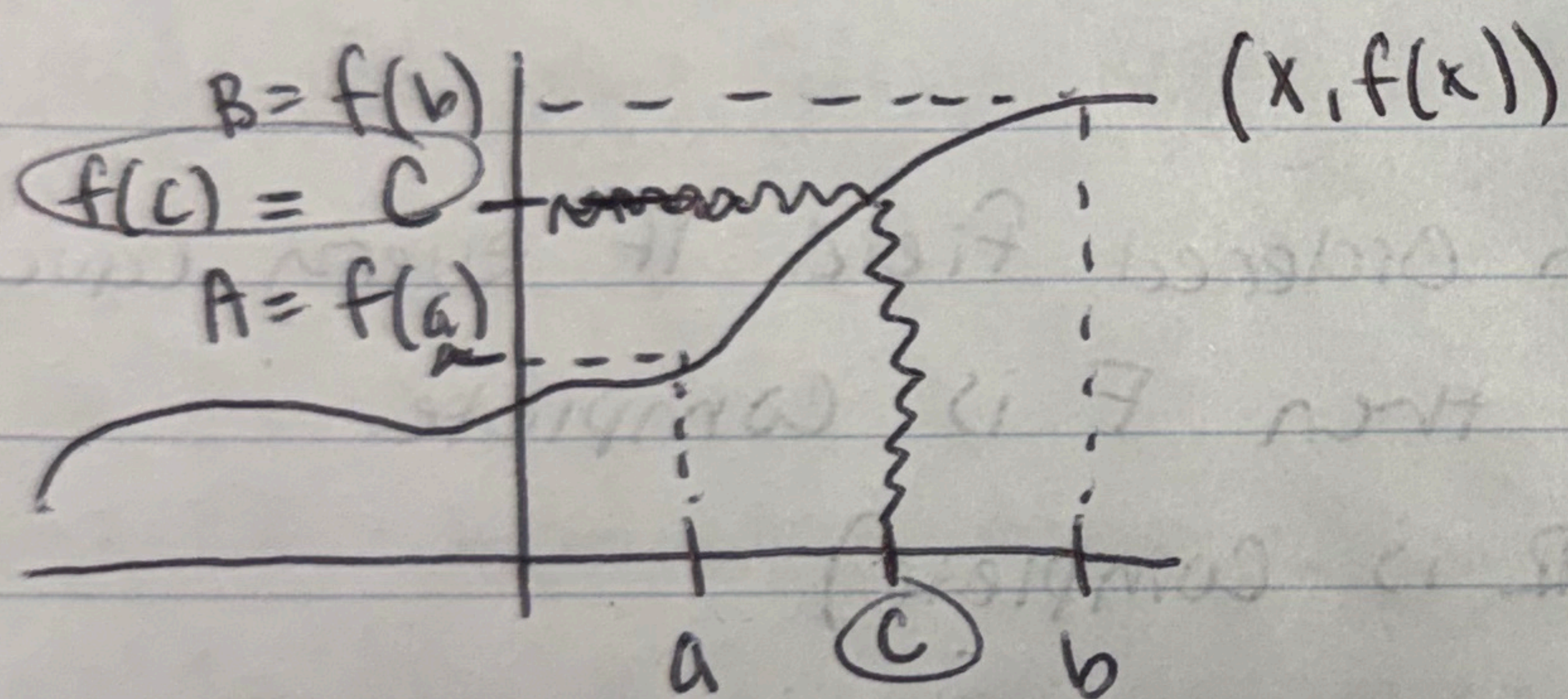
ex) $f: \mathbb{Q} \rightarrow \mathbb{Q}$ given by $f(r) = r$

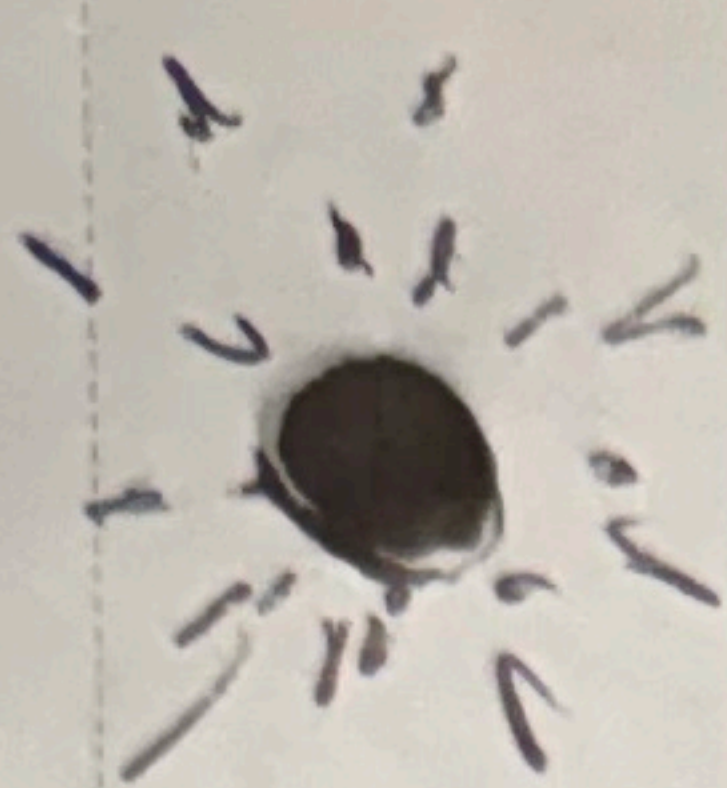
$g(x) = x^2$ from $\mathbb{Q} \rightarrow \mathbb{Q}$

$h(x) = \sqrt{x}$ as a function from $\mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ (not defined on $\mathbb{Q}$)

Thm. 12 Intermediate Value Thm.

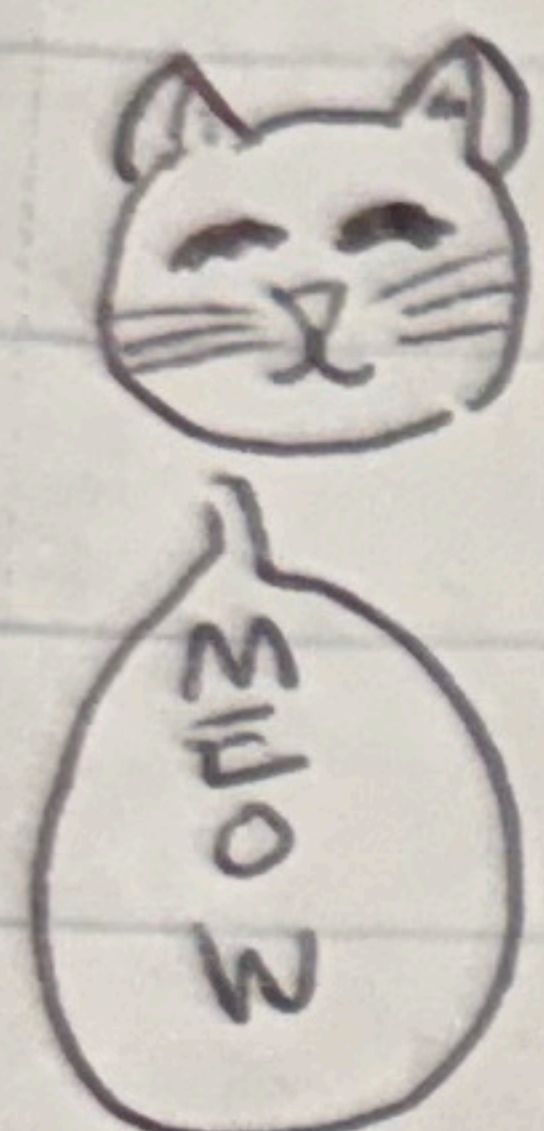
Let F be a complete ordered field that contains $\mathbb{Q}$ as a subfield. Let $[a, b] \subset F$ where $a < b$. Suppose $f: [a, b] \rightarrow F$ is continuous. If $C \in F$ is any value between $A = f(a)$ & $B = f(b)$, then $\exists$ an element $c \in [a, b]: f(c) = C$.





→ Consider $S \subset \mathbb{Q}$ defined by $S = \{r \in \mathbb{Q} : r^2 \leq 2\}$. It is bounded above by 2. Does it have a supremum? No

Ch. 8 [Thm. 6.0] → [Thm. 1.9] Ch. 9



Let S be a nonempty subset of an ordered Archimedean field F . If m is the supremum of S , m is the limit of a sequence (a_i) in S . Further, m is the limit of a sequence (b_i) not in S . If $a < m$, we can assume each $a_i \in [a, m]$. If $b > m$, " " $b_i \in [m, b]$.

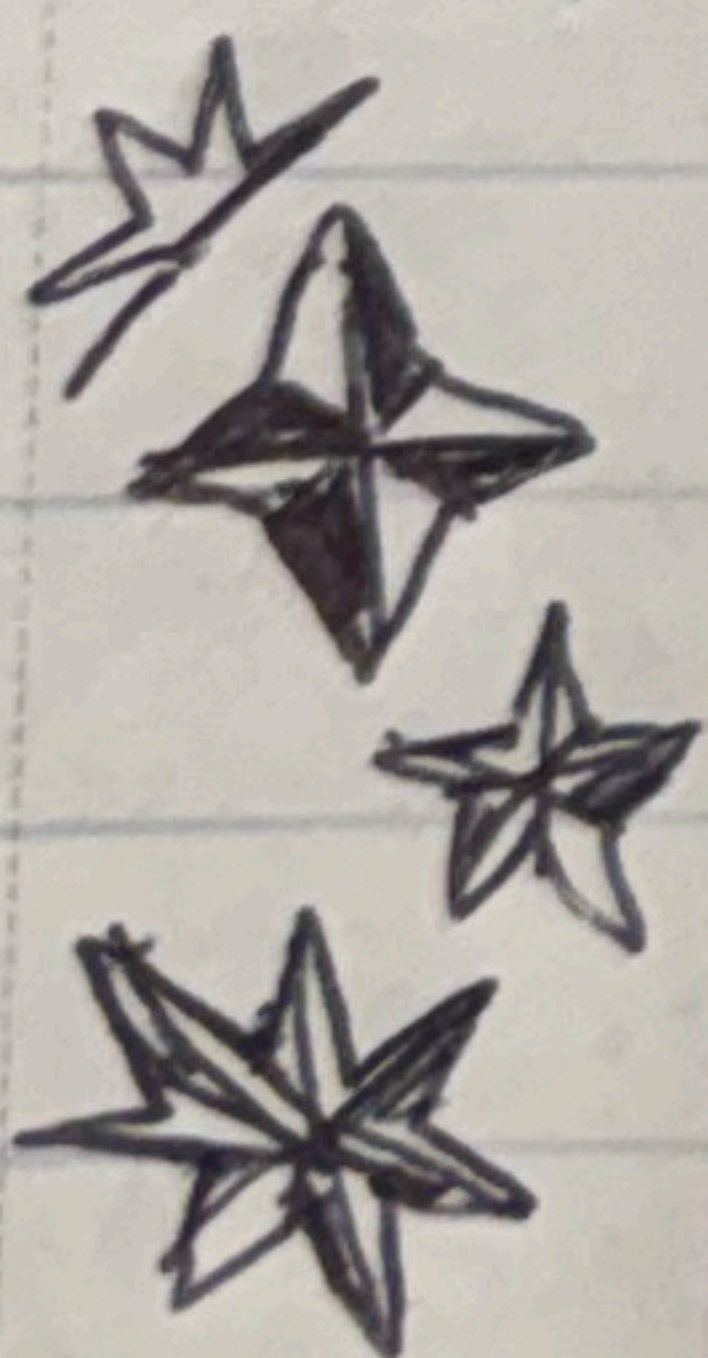
[Thm. 1.8] Ch. 9

Let F be an Archimedean Ordered Field. If every Cauchy sequence in F converges, then F is complete.
(This will end up showing $\mathbb{R}$ is complete)

Def. 2, Ch. 9

Continuity:

Let S be a subset of an ordered field F . Then a function $f: S \rightarrow F$ is said to be continuous on S if $\forall$ convergent sequences (a_i) with terms & limit in S , the sequence $(f(a_i))$ also converges. The $\lim_{i \rightarrow \infty} f(a_i) = f(\lim_{i \rightarrow \infty} a_i)$.
"f respects convergent sequences"



* how to prove stff converges/diverges

pf

step 1

Let $\epsilon > 0$, since $\mathbb{R}$ is Archimedean, $\exists n \in \mathbb{N}$ st $1/n < \epsilon$. Choose $N = n$ & suppose $i, j \geq N$. We have

$$\left| \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j}{10^j} \right| = \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j (0 \dots 0)}{10^j} \quad \{i, j \text{ zeros}\}$$

$$= \frac{d_{i+1} \dots d_j}{10^i}$$

$$< 10^{i-j} / 10^i = 1/10^j \leq 1/10^N = 1/n < \epsilon$$

Thus (a_i) is Cauchy, so $[(a_i)] \in \mathbb{R}$.

step 2

do on your own ☺

Note: $2 \in \mathbb{R}$ is the equivalence class of the constant sequence $(2)_i$.

Ch. 8

Def. 14 Supremum (least upper bound)

Let S be a non-empty subset of an ordered field F .

The supremum m of S is the least upper bound. I.e.

$m \geq x \forall x \in S$ & if $B \in F$ is such that

$B \geq x \forall x \in S$, then $m \leq B$.

The Infimum (greatest lower bound) has a similar def.

Ch. 9

Def. 1 Completeness

Let F be an ordered field. We say F is complete if every non-empty subset S which is bounded from above has a supremum.

ex →



Tuesday, May 5

Final:

- questions on what we've been doing recently
- epsilon "falling" from the sky
- convergence questions *
- induction (I think)

Final

Big Idea → "The Real #'s Have Enough Stuff" — $\mathbb{R}$ is complete

1. warm up: prove $\sqrt{2} \in \mathbb{R}$

step 1. show the sequence a_i , $i \in \mathbb{N}$ defined by

$$a_i = \frac{d_0 \cdots d_i}{10^i} \text{ with } \left(\frac{d_0 \cdots d_i}{10^i} \right) \leq 2 \text{ but } \left(\frac{d_0 \cdots d_i}{10^i} + \frac{1}{10^i} \right)^2 > 2 \text{ is}$$

Cauchy,

step 2. show $\lim_{i \rightarrow \infty} a_i^2 = 2$ so that $\sqrt{2} \in \mathbb{R}$ is $[a_i]$

$$0 \leq d_0 \leq 9$$

$$a_0 = d_0/10^0 = d_0$$

$$a_0^2 = d_0^2 < 2 \text{ but } \left(d_0 + \frac{1}{10^0} \right)^2 > 2$$

$$1^2 = 1 < 2 \quad \left(1 + \frac{1}{10^0} \right)^2 = (2)^2 = 4 > 2$$

Scratch

WTS: $a_i - a_j < \epsilon$ for i, j sufficiently big.

$$\left| \frac{d_0 \cdots d_i}{10^i} - \frac{d_0 \cdots d_j}{10^j} \right| \text{ if } i, j \text{ zeros}$$

$$= \left| \frac{d_0 \cdots d_i}{10^i} - \frac{d_0 \cdots d_j (0 \cdots 0)}{10^j} \right|$$

$$= \frac{d_{j+1} \cdots d_i}{10^i} < \frac{10^{i-j}}{10^j} = \frac{1}{10^j}$$

$$|a_3 - a_1| = \left| \frac{1414}{10^3} - \frac{14}{10^1} \right| = \frac{1414}{10^3} - \frac{1400}{10^3} = \frac{14}{10^3} < \frac{10^{3-1}}{10^3} = \frac{14}{100} < \frac{1}{100}$$